

GEARS: Seeing Geometry, Diffusing Actions for Zero-Shot Sim-to-Real Dexterous Manipulation

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Abstract. Simulation provides a scalable and annotation-rich data source for training dexterous manipulation policies, yet transferring such policies to the real world remains challenging. The sim-to-real gap manifests along two coupled axes: visual domain shifts cause perception modules trained on synthetic images to fail under real-world appearance variations, while dynamics mismatch destabilizes control during contact-rich manipulation. We present GEARS, a unified framework that addresses both axes by coupling geometric visual representations with generative action modeling. For perception, we learn texture-invariant geometric features from monocular RGB by fine-tuning a visual foundation model with joint depth and surface normal prediction using privileged simulation labels, enforced by a cross-task consistency constraint. For control, we train an RL expert with an efficient particle-based automatic domain randomization strategy that steers training toward difficult regions, and distill its behavior into a diffusion policy conditioned on the learned geometric features. A key insight is that these two components are mutually reinforcing: the geometric features provide a stable, domain-invariant conditioning space that simplifies the denoising problem, while the diffusion formulation captures the multimodal action distributions induced by domain randomization and produces temporally coherent trajectories. We evaluate GEARS on zero-shot sim-to-real dexterous manipulation in chemistry laboratory environments, where transparent glassware and reflective surfaces pose significant perceptual challenges. GEARS achieves 67.5% average success across four tasks, outperforming the strongest baseline by 25 points.

Keywords: Sim-to-Real Transfer · Dexterous Manipulation · Geometric Representation Learning

1 Introduction

Learning generalizable manipulation skills is a central goal of embodied intelligence [2, 14, 23, 43]. Training such skills requires large-scale and diverse data covering a wide range of objects, tasks, and physical conditions. Among available data sources, simulation is especially appealing: it can generate data at low cost [36] and provides privileged geometric annotations such as dense depth and surface normals that are difficult to obtain in the physical world. However, a persistent gap between synthetic and real environments prevents these advantages